

```

Sxh = sx Cos[θ] - sy Sin[θ];
Sxxh = sxx Cos[θ]2 - 2 sxy Cos[θ] Sin[θ] + sy Sin[θ]2;
SSxxh = ExpandAll[Sxxh - Sxh2];

Syh = sx Sin[θ] + sy Cos[θ];
Syyh = sy Sin[θ]2 + 2 sxy Sin[θ] Cos[θ] + sxx Sin[θ]2;
SSyyh = ExpandAll[Syyh - Syh2];

Sxyh = sxy Cos[θ]2 + sxx Cos[θ] Sin[θ] - sy Sin[θ] Cos[θ] - sxy Sin[θ]2;
SSxyh = ExpandAll[Sxyh - Sxh Syh];

R2 = FullSimplify[SSxyh2] / FullSimplify[(SSxxh SSyyh)]

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$$\frac{\left((sxy - sx sy) \cos[2\theta] + (-sx^2 + sxx + sy^2 - syy) \cos[\theta] \sin[\theta] \right)^2}{\left((sx^2 - sxx) \cos^2[\theta] + (sy^2 - syy) \sin^2[\theta] + (sxy - sx sy) \sin[2\theta] \right) \left((sy^2 - syy) \cos^2[\theta] + (sx^2 - sxx) \sin^2[\theta] + (-sxy + sx sy) \sin[2\theta] \right)}$$

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sx = 5.5;
sy = 14.9;
sxx = 38.5;
syy = 260.542;
sxy = 98.31;
Plot[R2, {θ, -π/2, π/2}]

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